Need for EMI shielding

Electromagnetic radiation passes through the air and can interfere with the proper operation of various types of equipment. This is one of the basic reasons that EMI and electromagnetic compatibility (EMC) testings are performed. The circuit should be designed to tolerate EMI from the environment and also minimise the amount of electromagnetic radiation that it generates by providing sufficient electromagnetic shielding.

In the design process of any space vehicle, especially in the aerospace industry, electromagnetic shielding plays an important role. If proper protection is not implemented, electromagnetic fields from various devices may have a tremendous effect on each other. It is mainly required to provide an effective shielding to the buildings containing power transformers and other electronic facilities, which will radiate electromagnetic waves into the environment.

Important areas of EMI shielding

The rapid development of electronic industries and the widespread use of electronics equipment in space, automation, communication, computations, biomedical, and other purposes has led to many EMI problems for the designers as their system or sub-systems operate in close proximity. If designers do not follow the EMI control technique to meet the EMC requirement during the design stage itself, it is likely to become a more severe situation in the future. The prime objective of the design should be elimination or suppression of EMI.

Shielding is typically applied to enclosures to isolate electrical devices and cables from the outside world. Electromagnetic shielding that blocks radiofrequency (RF) electromagnetic radiation is also known as RF shielding. The significance of EMI shielding relates to the high demand of today's society on the reliability of electronics and the rapid growth of radio frequency radiation sources.

The following areas need EMI/EMC concern:

- Defence and space
- Aerospace
- Communication
- Automobile
- Electronics
- Healthcare
- Materials
- Energy

Scope for EMI shielding

EMI/EMC first came into view in the 1940s and 1950s. At that time, problems were more focused on motor noises that were conducted over power lines and into sensitive equipment.

From the 1960s, EMI/EMC was primarily of interest to the military to ensure electronic systems in military services which do not interfere with each other. Without proper EMC consideration, accidents such as radar emissions causing inadvertent weapons release, or emissions from electronic devices causing navigation systems failure may occur and degrade the defence capability of the weapon systems.

During the 1970s and 1980s, the computer industry flourished, and interferences from computing devices became a significant problem to broadcast television and radio reception. This made the government to decide to regulate the amount of electromagnetic emissions from products in this industry. The Federal Communications Commission (FCC) of the USA issued a set of rules to govern the amount of emissions from any type of computing device. Similarly, European and other governments began to control emissions from electronic and computing devices.

Contribution of India

From the 1990s, the concern over EMI/EMC has been found to broaden dramatically; in fact, most of the countries have now instituted import controls requiring that EMI/EMC regulations be met before products can be imported into the country. The overall compatibility of all devices and equipment must coexist harmoniously in the overall electromagnetic environment. Emission, susceptibility to electrostatic discharges, susceptibility to emission either from radiated or conducted media are controlled.

This control is no longer just limited to computers. Now any product that may potentially radiate EMI, or that could be susceptible to other emissions, must be carefully tested. Products with no previous need for EMI/EMC control must now comply with the regulations.

The task of developing standards for the control of EMI began with the formation of International Special Committee on Radio Interference (CISPR), and a special committee under the sponsorship of the International Electrotechnical Commission (IEC).

Indian standards (IS) on EMC are adopted by the Bureau of Indian Standards (BIS) after the draft finalised by the Electromagnetic Compatibility Sectional Committee (ECSC) is approved by the Electronics and Telecommunication Division Council. The IS on EMC is being revised to align with the different CISPR and IEC publications. BIS is a statutory institution established under the Bureau of Indian Standards Act 1986 to promote the harmonious development of the activities such as standardisation, marking and certifying quality products, and attending to connected matters in the country.

EMC laboratories in India

The research and development (R&D) in the area of EMI/EMC is actively progressing in India with support from the defence, space, atomic energy, and labs funded by Ministry of Electronics and Information Technology (MeitY), Council of Scientific and Industrial Research (CSIR), and other agencies. The government labs are also backed by various private recognised labs and companies for testing and certifications. The academic institutes are also contributing their best for the development of the domain area.

Professional bodies like societies for EMC engineers (India) bring together all researchers, engineers, scientists, industries, business entities, and students to a common forum through the mega international event like 'INCEMIC' to discuss and deliberate the issues, developments, and future prospects in this field. Defence Research and Development Organisation (DRDO) labs play a leading role in this and have successfully conducted this event for many years. EFY